

COURSE GUIDE · GRADES K-5, MIXED AGE

Kitchen STEAM



Science, Technology, Engineering, Art and Mathematics through cooking,
food exploration and the table

Forty-six activity cards · a course guide · a food journal · grades K-5, mixed age

Everything in this guide is designed so that a child who never tastes anything completes the whole course, and so that nobody in the room can tell which child that is.

The one page, if you read nothing else

Post this. Give it to every adult who comes into the room, including substitutes and parent helpers, **before** they arrive.

1 Nobody has to taste anything. Ever.

Cooking, measuring, smelling, describing, drawing, serving and clearing are full participation in this class — not a lesser version of it. This is stated in the letter home before the first session, so that every family knows it and no child has to ask.

2 No praise for eating, and no reward for tasting.

Praise for consumption is pressure wearing a friendly face, and in a group it converts a private moment into a public performance that every non-taster has just failed. Comment on the work instead: *your dice were really even, you found three words nobody else had*. Never on what or how much anyone ate.

3 “Not for me today” is a complete sentence.

It needs no reason and no follow-up. The whole of the adult response is *okay*, and then a change of subject — no pause, no second glance, no *are you sure*. Every place setting has a small plate for food a person does not want. Every setting, including the adults'.

4 Nobody comments on what anyone else eats.

This is taught as a rule in week one and enforced all term, because peer policing is the real risk in a group. It protects the child who eats little, the child whose food from home looks unfamiliar, the child whose family eats differently, and the child whose lunchbox is thin some weeks. Your job is your own plate.

5 Every ingredient is named, in advance, always.

The full term's recipes and ingredient lists go to every family before the first session. Nothing is ever hidden in a dish as a pedagogical surprise. A hidden ingredient revealed afterwards is a trust rupture, and for an anxious eater it can generalize into avoidance of the dish, the room and you.

6 We describe. We do not judge.

Yuck, gross and pulled faces are out — not because they are rude, but because refusal is contagious in a group in a way that description is not. The redirect is always the same and always calm: *give me a word for what it is actually like*.

7 There are no good foods and no bad foods.

No sorting into healthy and unhealthy, no calories, no talk about bodies, weight or burning anything off, no *this is fuel for you*. Children are measurably less likely to try a food that has been labeled healthy. Teach what food *does* and where it comes from — which is the STEAM content anyway.

8 Everyone has a named job, and the jobs rotate.

Reader, measurer, mixer, timekeeper, sensory reporter, station chief, photographer, server. Jobs are assigned by rotation, not by who eats what, and the sensory reporter — the highest-status role that requires no tasting — must never become one particular child's post.

What this class is, and what it is not

This is a STEAM course that uses food as its laboratory. The children learn chemistry, material science, engineering design, drawing, printmaking, composition, statistics, measurement, sensory science and something about where food comes from and who grows it. They also cook, and they eat if they want to.

It is not a nutrition course, it is not a healthy-eating intervention, and it is emphatically not feeding therapy. Those boundaries are not modesty — they are what makes the class safe to run for a full mixed-age group without knowing which children in the room have a feeding disorder, an eating disorder, a sensory processing difference, a food allergy, a restricted household budget, or simply a strong opinion about mushrooms.

The design principle throughout is **universal design rather than accommodation**. Nothing here is a special arrangement made for a particular child. The no-thank-you plate is on every setting; the components are always served separately; headphones are available to anyone; the recipes go to every family in advance. A child who needs those things gets them without being identified, and every other child benefits from a calmer, more predictable room.

The single most useful thing a school class can offer a child who is working on food with a clinician outside school is **a predictable, pressure-free environment in which food is handled constantly and nothing is demanded**. That is a real contribution and it is entirely within a teacher's remit. Running a private feeding program is not.

Why the assignment is description, not tasting

The spine of this course is sensory food education — the approach developed as the Sapere method in Finland and Sweden, now embedded in several national early-years frameworks. Its defining move is to shift children from *I like it / I don't like it* to description: what does it look, sound, smell, feel and taste like, in words.

There is decent evidence behind it. Studies of sensory education with eight- to twelve-year-olds have found reduced food neophobia and increased willingness to try unfamiliar foods, and classroom studies have found that groups doing sensory work do not show the drift toward produce avoidance that ordinary groups do — refusal spreads through a group, and description appears to interrupt it.

But the structural argument matters more than the evidence, and it is the reason this course is built the way it is. **If the assignment is a description, then a child who only looked and smelled has completed it.** There is no moment at which some children succeed and others fall short, nothing has to be adapted for anybody, and no accommodation is visible to the room. That is what universal design looks like when it works.

It also happens to produce better teaching. A class that can say *brittle, fibrous, sharp, squeaky* and *fragrant* is doing vocabulary work, observation work and the beginnings of sensory science, none of which a class that can only say *nice* and *yuck* can do.

Eight ways to meet a food — and why it is a wheel

Feeding therapists work with a long graduated hierarchy. The best-known runs to more than thirty steps across six bands: tolerating a food in the room, then at the table, then nearer; interacting with it; smelling it; touching it in a progression that works inward from fingertip to hand to arm to face to lips; tasting; and finally eating. Roughly the first half involves no contact with the mouth at all.

The band called *interacts with* is worth quoting almost in full, because it is a description of a cooking class: helping prepare and set up, using a utensil to stir or pour for other people, serving food onto one's own plate, using tools to move food around one's own space. **A kitchen is already most of that ladder.** You do not need to add a tasting requirement in order to be doing real exposure work; you need to run a kitchen properly and then get out of the way.

Two deliberate departures from the clinical version. The class version is drawn as a **wheel and is not numbered**, because a ladder implies a top, a top implies falling short, and in a group that turns a private step into a public ranking. And **no record is kept of where any child is**. Building an individual child's hierarchy, choosing their next step, or setting a goal for them is clinical work that belongs to that child's own team.

The eight ways, as the children meet them: **be near it · look · listen · smell · move it · touch it · lip it · taste it**. All eight count. Any one of them is meeting the food. Card N02 teaches this in the first session and the wheel stays on the wall all term.

One more thing, and it overrides everything above. **If a child's outside team already uses step language, the class adopts theirs** — for everyone, on the wall, from week one. The wheel is not a fixed artefact and the class version is the one that yields. Ask about this on the intake form, before the first session, not after.

Where your job stops

Be firm about this boundary. It protects the child and it protects you.

Not yours: diagnosing, screening for, or naming any eating or feeding disorder in a child. Building or advancing an individual child's exposure hierarchy or food chain. Setting per-child tasting goals. Keeping a per-child record of what anyone ate. Any decision about a child with swallowing or choking risk — that requires a speech and language therapist and is a formal clinical process. Nutritional advice, and anything at all about weight.

Yours: designing a room where the lower half of that ladder happens constantly and unremarkably. Teaching sensory science, cooking skill and food culture as real curriculum. Holding the norms for every adult and child who comes through the door. And noticing — then passing to the designated coordinator, not acting on — signals such as a child's range of accepted foods visibly narrowing, distress escalating rather than settling, or a child beginning to avoid the class. Refer; do not intervene.

If a child's family has an outside feeding therapist, occupational therapist or speech and language therapist, contact goes through one named person at the school — usually the nurse, counsellor or case manager — and only with a signed release from the family. What you ask for is *strategies and step language*, so that you use the same words the child already knows. What you do not ask for, and do not record, is the diagnosis. You do not need it.

Language

Most of what makes this class safe or unsafe is said out loud in the moment, usually by a tired adult with something in the oven. The list opposite is worth photocopying and taping inside a cupboard door.

One phrase deserves particular attention, because it is well-meant and it is a trap: *brave*. Telling a child they are brave for tasting something implies the food is frightening and implies that the child who did not taste is not brave. In a group, prefer role language to virtue language — *you were our smell describer today* rather than *you were so brave*.

The second trap is your own self-talk. An offhand *I really shouldn't* or *I'm being so bad* from an adult is curriculum, and children absorb it faster than anything on the board.

SAY THIS INSTEAD

Instead of “Just try one bite.”	Say nothing. Offer a job: “Do you want to be our describer for this one?”
Instead of “Great job trying it!”	“You found a word nobody else had.” Comment on the work, never the eating.
Instead of “You were so brave.”	“You were our smell describer today.” Role language, not virtue language.
Instead of “Are you sure? You liked it last time.”	“Okay.” Then change the subject.
Instead of “Look, Maya tried it!”	Nothing. Never recruit a peer as a motivator.
Instead of “It’s healthy, it’s good for you.”	“It’s got a snap to it. Listen.” Function and sensory, never virtue.
Instead of “Don’t say yuck, that’s rude.”	“Give me a word for what it’s actually like.”
Instead of “Everyone else is having some.”	Nothing. Comparison is the thing this room does not do.
Instead of “Finish that and you can have...”	Nothing is contingent on eating. No food is a reward or a consequence.
Instead of “Guess what was hidden in it!”	Every ingredient named before it goes in. Always, without exception.

A room with five grades in it

Mixed-age is an advantage here, not a compromise, but it has to be structured or the ten-year-olds will do everything and the six-year-olds will watch.

Stations of five or six, each with one adult, each making something different, with small taster portions shared across stations at the end. Small portions and high variety is a quietly excellent low-pressure design: nobody is committed to a full plate of anything.

Named jobs, rotated every session, assigned by rotation and not by age — reader, measurer, mixer, timekeeper, sensory reporter, station chief, photographer, server. The sensory reporter is the inclusion engine of the whole course: high status, genuinely difficult, requires no tasting. Rotate it deliberately so it never becomes the job for the child who does not eat.

Pairing. For working together on a task, a gap of less than three years usually pairs better than a wider one — a fifth-grader and a third-grader collaborate, where a fifth-grader and a kindergartener tend to slide into doing-and-watching. Wide pairs are excellent for reading aloud, mentoring and user-testing. State the rule out loud and repeat it: *your buddy's hands do the work, your words help.*

Give the youngest children real jobs with real consequences. Several cards in this deck deliberately hand the most important role to the youngest — the user tester who cannot read the recipe, the silent tester of a prototype, the person who goes to the office for the real roll number. That is not kindness; it is where the best data comes from.

Knives, heat and hygiene

Both real hazards are better handled by graduated responsibility than by prohibition. A child who has never been near a knife at eight is not safer at twelve, only less practiced.

K-1: tearing, mashing, whisking, measuring, pressing; crinkle cutters, table knives and nylon lettuce knives on soft foods. **Grades 2-3:** small serrated children's knives with a bridge or claw grip on stable foods; graters with guards; peelers, stroking away from the body. **Grades 4-5:** small chef's knives on a board anchored with a damp towel, under supervision, and stovetop work with an adult at that station.

Heat is adult-gated at every age, with no exceptions. One adult per live heat source. Ovens are opened by adults. Boiling water and hot fat are carried by adults, and never across a room where children are moving. Molten sugar and deep fat do not belong in this course at all.

Name the hazard out loud before it is live — *the pan handle is turned in so nobody catches it with a sleeve* — because children who hear hazards named learn to name them, which is the skill that actually transfers.

Standing rules: hands washed before and during; hair tied; anything dropped is discarded; no sharing of utensils or plates; separate boards and equipment for raw meat if it appears at all; a single agreed stop signal that means all hands still. And one that is less obvious — **do not clean children as they work**. Constant wiping interrupts tactile exploration and communicates that the food is contaminating. Put warm soapy water within reach and let them wash when they choose.

Allergies, restriction, culture and cost

Design the menu around the roster rather than managing around it. If a child in the group has a serious allergy, the straightforward course is to keep that allergen out of the classroom entirely rather than run a containment operation every week — that is what public-health guidance for schools recommends and it is far less fragile than the alternative.

Whatever the restriction, the child should be doing the same science, not a diminished version. Emulsions work with aquafaba. Leavening works in gluten-free batter. Gels work with agar. Where a substitution genuinely changes the result — agar is not affected by the enzyme in fresh pineapple, for instance — make that the control condition and the child's version becomes the more interesting one.

Religious and cultural practice is asked about on the intake form and planned around quietly, not negotiated in the room. **Cost** is a live consideration too: never ask families to send in ingredients, never assign homework that requires buying food, and never ask a child to record what they ate at home. Any of those exposes a household's finances to the class.

Two allergen habits worth building. Everything the class sends out of the room carries a complete ingredient list. And no food is ever used as a reward or withheld as a consequence — the allergy rationale and the eating-disorder rationale point the same way here.

Allergies, restriction, culture and cost — continued

One standing allergen rule, rather than forty-six. Check the roster against every card's materials before the term is planned, not the week before the session. Wheat and celiac disease deserve particular attention, because airborne flour — not eating it — is the hazard in the dough cards, and a child with either cannot safely be in the room. Ask about animal-derived *materials* as well as foods: gelatin is usually pork or beef and appears in a card where agar does the same job.

Any dish not in the term pack goes to every family, with its complete ingredient list, at least a full week before it is cooked. If that notice cannot be given, the dish is not cooked. This applies to the family dishes in C01, to anything cooked for a recipient in C05, and to every substitution.

Food that becomes a specimen. Several cards deliberately turn food into props, data and evidence and then throw it away. Say what will happen to the food at the start of the card, before it is handled — surprise disposal is worse than planned disposal. Route it somewhere real where you can: trimmings to the stock pot, whole unhandled produce back to the kitchen or offered at the door, specimens to a compost the class can see. And notice who might be watching. For a child whose household counts food carefully, a discarded tray is not a neutral event, and no explanation afterwards repairs it.

The food journal — and five things never to do with it

The journal is the child's, not the school's. It is where they draw, describe, measure, graph and record what they thought, and it is designed so that **a child who never tastes anything can complete every page of it** — every rating page carries a box for *I didn't taste this one*, and the description work is all sensory rather than hedonic.

It is deliberately built around **intensity rather than liking**. The main scales are *how crunchy*, *how strong a smell*, *how sweet* — scales where there is no right answer and no virtue in a high score. The preference question is smaller, optional, and always attached to a *why* written in sensory words.

Never collect it, grade it, compare it, or display it. The moment a journal is assessed it becomes a compliance document. If you want to see a page, ask the child to show you, and accept no.

Never record quantity eaten, number of new foods tried, or anything that accumulates across weeks into a per-child progress chart. Intake logging is contraindicated for children with eating disorders, and there is no pedagogical benefit it offers that sensory recording does not offer better.

Never send a food diary home. Do not ask children to record what they ate at home, for the week, or at all. It exposes household food insecurity to the class and it teaches that food is something to be audited.

Never make a wall display of who tried what. Class-level anonymous results are excellent and card M05 is built around them; individual tallies are not.

Never carry a rating forward as evidence. A rating is one child's impression on one day. If a child rates something 1 in September and 4 in November, that is theirs to notice, not yours to point out.

Assessment, without counting tastes

If your assessment, your display board or your parent newsletter counts tastes, everything else in this document is decoration. Make the unit of success **participation in a role**.

The role record. Over ten or sixteen weeks, did each child hold every job at least once — reader, measurer, timekeeper, sensory reporter, station chief, server? That is a legitimate, checkable record of participation and it correlates with nothing about eating.

Vocabulary growth. The word wall from card N01 is a class artefact that visibly grows. Photograph it in week one and week ten. The journals are not evidence of anything and are not looked at for this or any other purpose.

The redesign paragraph. After any engineering card: *my first design failed because ____, I changed ____, the result was ____*. Three blanks, the whole design cycle, and the most transferable artefact in the course.

Prediction slips. Before an experiment: *I think ____ will happen because ____*. The *because* half is what you are assessing.

Teach it to someone smaller. Mixed-age makes this native. A fifth-grader who can explain gluten to a six-year-old without using a word the six-year-old does not have understands gluten.

The letter home — a draft you can send

Send this before the first session, with the full term's recipes and ingredient lists attached. The third paragraph is the highest-leverage text in this entire document: it tells a family with a worried eater that their child is safe, and it tells every other family what the norm is — which pre-empts pressure arriving from home.

Dear families,

This term your child is joining Kitchen STEAM, a cooking and food-exploration class where we do science, engineering, art and math using food as our laboratory. We will make natural dyes from onion skins, print with vegetables, build and test pasta shapes, count the spirals on a pineapple, draw a pepper's cross-section, film a recipe frame by frame, and cook things from families in our own school community.

No child is ever required to taste anything in this class. Cooking, measuring, smelling, describing, drawing, photographing, serving and clearing all count as full participation — they are not a lesser option, and children who never taste a thing can and do complete every part of the course. We never praise or reward children for eating, and we never comment on what anyone has eaten. If your child says “not for me today”, our whole reply is “okay”.

Every ingredient is named in advance. The full term's recipes and ingredient lists are attached, so you and your child can look at them together before each session. We never hide or disguise an ingredient in a dish, and we will tell you in advance if anything changes.

If we cook anything that was not in the term pack — a family's dish, or something the class chooses to cook for somebody — you will have it, with its full ingredient list, at least a week beforehand. If we cannot give you that much notice, we do not cook it.

Please fill in the short form attached so we can plan around allergies, medical diets, and your family's practices. There is an open question at the end — “anything you'd like us not to do?” — and we read every answer to it carefully.

Your child will keep a food journal. It belongs to them. We do not collect it, grade it, or compare it, and we never ask children to record what they eat at home.

You are welcome in the kitchen. If there is a dish that matters in your family and you would be willing to come and cook it with us, or to tell us about it, we would be glad — and equally, there is no expectation at all.

The family intake form

ASK

- Allergies, intolerances and medically prescribed diets (with the school's usual medical verification).
- Cultural, religious or family dietary practices we should plan around.
- Foods your child reliably eats that you would be glad to see in class.
- Anything about smells, textures, noise or the kitchen environment we should know.
- Would your child like the option of a component prepared in advance, or of bringing their own?
- Is your child already using step language for trying foods that we should use instead of ours? (We will change the class's words, for everybody.)
- Is there an outside provider you would like us to coordinate with? (If so, we will send a release form.)
- Who should receive advance notice of recipes, and how?
- **Anything you would like us NOT to do?** — the highest-yield question on the form.

DO NOT ASK

- A diagnosis, or any label identifying your child as having a feeding or eating disorder. We do not need it, and collecting it creates a record with disclosure risk.
- Anything about weight, height, growth or “concerns about your child’s weight”.
- “How many foods does your child eat?” or “Is your child a picky eater?” — deficit framing, and these families have usually been blamed for this already.
- A food diary, a 24-hour recall, or any record of what the child eats at home.
- Anything implying that tasting is expected. State the opposite, explicitly, in writing.

Brief every adult. Photocopy this page.

An untrained volunteer saying “just try one bite” is the most likely way this course fails. Hand this to every aide, substitute, student teacher and parent helper **before** the session, not during it.

KITCHEN STEAM • TEN RULES FOR ANYONE HELPING IN THIS ROOM

1. **Nobody has to taste anything.** Not one bite, not a lick, not “just try it”. Cooking, smelling, describing and serving are full participation.
2. **Never comment on what a child ate** — not how much, not what, not whether. Not even approvingly. Especially not approvingly.
3. **If a child says “not for me today”, say “okay” and move on.** No pause, no second look, no “are you sure”.
4. **No rewards for tasting** and no food used as a reward or withheld as a consequence.
5. **Never point out one child's eating to another.** No “look, she tried it”.
6. **Never say a food is healthy, bad, junk, or good for you.** Describe it instead — crunchy, sharp, fragrant.
7. **Watch your own words about yourself.** “I shouldn’t” and “I’m being bad” are heard by everyone.
8. **Every ingredient is named out loud.** Never hide anything in a dish, even as a joke.
9. **If a child says “yuck”,** the reply is “give me a word for what it's actually like”.
10. **Heat is yours, not theirs.** One adult per heat source. Ovens opened by adults only.
11. **Never mention a child's eating to another adult either** — not to a colleague, and not to that child's parent at pickup. “She ate the soup!” is the same comment, just delivered later.

Sequence A — the ten-week after-school club

TEN SESSIONS • 75 MINUTES • ONE MIXED K-5 GROUP • REAL KITCHEN

A complete arc in ten weeks. Week one sets the norms and hands out the journals; the middle weeks run long-lead projects — dyeing, printmaking, a sourdough of sorts — that carry across sessions; the last two weeks turn outward, to the families and to somebody who will be glad of what the class made. Cards appear in the order they are run.

1

The kitchen we are going to be

N02 Eight Ways to Meet a Food • N01 The Taster's Alphabet

Norms, the agreement, the wheel on the wall, journals handed out. Do not cook anything ambitious. The room's character is set this week and nothing else matters as much.

2

Looking and listening

N03 Listen to It First • A02 A Color Wheel You Can Eat

Two sessions with no tasting expectation at all, which is deliberate. The word wall starts filling and the sound-guessing game does more for group cohesion than anything else in the term.

3

Color, and where it comes from

A01 The Color in the Peel — session one • S06 Four Families of Color

Dye baths simmer while the pigment families are tested. The two cards explain each other: the pigments that leach into cooking water are exactly the ones that make good dyes.

4

Color, continued

A01 The Color in the Peel — session two • A03 The Okra Is a Stamp

Mordanted cloth comes out, the pH shift happens, and the fade test goes on the windowsill. Printmaking starts on the dyed paper.

5

Bread as a material

E04 What Kneading Actually Does • A07 Dough Is a Material

The windowpane test and the gluten washing, then sculpture in the same material. A long, physical, floury session — schedule the clean-up.

6

Measuring, and the people who had none

M06 A Pinch, a Handful, a Gram • T05 Before Anyone Owned a Thermometer

The class's own spread of handfuls makes the argument for standardization. Low mess, high arithmetic — a good week after the bread.

7

Design and test

E02 Which Shape Holds the Sauce? • A05 The Plate Is a Frame

A genuine quantitative engineering test in the first half and composition in the second. Send the interview sheets from T02 home this week.

8

The dish that traveled

T02 Ask, Then Stop Talking • C01 The Dish That Traveled

Interviews come back; the map goes up; one or two family dishes get cooked with the family present if they want to be. This is usually the week the class turns into something.

9

For somebody else

C05 Cook It for Someone Else • A10 What Shall We Call It?

Cook, package, label completely, write the notes, deliver in person. Naming and menu design run alongside for whoever is not at the stove.

10

Family night

N05 A Table Where Everyone Fits • A06 Where Is the Light Coming From?

The class hosts. Dyed cloth, prints, drawings, the word wall, the photographs, the journals if children choose to show them. Components served separately; the norms posted at the door.

Cards not placed in either sequence are the bank, and they are not lesser — several are among the strongest in the deck. Swap them in when the kitchen is unavailable (M01, M02, M03, T02, T04, A10, E01 all run in an ordinary classroom), when the weather or the term throws a session, or when a card connects to something the children are already studying.

Sequence B — the sixteen-week semester course

SIXTEEN SESSIONS • 60-75 MINUTES • ONE MIXED K-5 GROUP • REAL KITCHEN

The same spine with room to breathe, and with a publication at the end. The arc runs senses → materials → chemistry → design → culture → publication, and the extra six weeks buy three things the club version cannot afford: the illustration and looking-closer cards, a proper statistics thread, and time to actually make the cookbook.

- | | |
|---|--|
| <div style="display: flex; align-items: center; margin-bottom: 10px;"> <div style="background-color: #4a4a8a; color: white; border-radius: 50%; width: 40px; height: 40px; display: flex; align-items: center; justify-content: center; margin-right: 10px;">1</div> <div> <p>The kitchen we are going to be</p> <p>N02 • N01</p> <p>Norms, agreement, journals.</p> </div> </div> <div style="display: flex; align-items: center; margin-bottom: 10px;"> <div style="background-color: #4a4a8a; color: white; border-radius: 50%; width: 40px; height: 40px; display: flex; align-items: center; justify-content: center; margin-right: 10px;">2</div> <div> <p>Listening and looking</p> <p>N03 • A02</p> <p>Sound and color. No tasting expectation.</p> </div> </div> <div style="display: flex; align-items: center; margin-bottom: 10px;"> <div style="background-color: #4a4a8a; color: white; border-radius: 50%; width: 40px; height: 40px; display: flex; align-items: center; justify-content: center; margin-right: 10px;">3</div> <div> <p>Closer than you have ever looked</p> <p>S07 • A04</p> <p>Magnification, then observational drawing. Quiet, precise, popular.</p> </div> </div> <div style="display: flex; align-items: center;"> <div style="background-color: #4a4a8a; color: white; border-radius: 50%; width: 40px; height: 40px; display: flex; align-items: center; justify-content: center; margin-right: 10px;">4</div> <div> <p>It changed and nobody added anything</p> <p>S04 • M05</p> <p>Enzymatic browning, then the first anonymous class vote.</p> </div> </div> | <div style="display: flex; align-items: center; margin-bottom: 10px;"> <div style="background-color: #4a4a8a; color: white; border-radius: 50%; width: 40px; height: 40px; display: flex; align-items: center; justify-content: center; margin-right: 10px;">5</div> <div> <p>Color out of the trash</p> <p>A01 — session one</p> <p>Dye baths, mordanting, tying.</p> </div> </div> <div style="display: flex; align-items: center; margin-bottom: 10px;"> <div style="background-color: #4a4a8a; color: white; border-radius: 50%; width: 40px; height: 40px; display: flex; align-items: center; justify-content: center; margin-right: 10px;">6</div> <div> <p>Color, continued</p> <p>A01 — session two • S06</p> <p>The pH shift, the fade test, the pigment families.</p> </div> </div> <div style="display: flex; align-items: center; margin-bottom: 10px;"> <div style="background-color: #4a4a8a; color: white; border-radius: 50%; width: 40px; height: 40px; display: flex; align-items: center; justify-content: center; margin-right: 10px;">7</div> <div> <p>Pattern and print</p> <p>A03 • A08</p> <p>Relief printing, then radial symmetry and repeat units.</p> </div> </div> <div style="display: flex; align-items: center;"> <div style="background-color: #4a4a8a; color: white; border-radius: 50%; width: 40px; height: 40px; display: flex; align-items: center; justify-content: center; margin-right: 10px;">8</div> <div> <p>Bread as a material</p> <p>E04 • A07</p> <p>Windowpane, gluten washing, sculpture.</p> </div> </div> |
|---|--|

Sequence B — the sixteen-week semester course — weeks 9 to 16

9

Water moving where you did not put it

S01 • S03

Osmosis, then the three gels. Both are prediction-heavy.

10

Measuring, and the people who had none

M06 • T05

Handfuls versus grams; the flour oven test.

11

Engineering with a number on it

E02 • N04

Pasta shapes tested by weight, then the triangle test. Six roles in N04 and only one of them involves a mouth — the coder who assigns the random numbers is the most important person in the room.

12

Whose tongue is this?

S05 • A08

Plant defenses and why people taste bitterness differently, paired with radial pattern — so the session's second half is led by children who are counting rather than tasting.

13

Ask, then stop talking

T02 • C04

Interview technique, then the chain of people behind one apple. Invite the school's own kitchen staff.

14

The dish that traveled

C01 • C02

Family dishes cooked with families; one grain across many kitchens.

15

Make the book

T04 • A10 • M02

Editing, naming, layout — and scaling a recipe for the whole school. The K-1 user test of the recipes is the best twenty minutes of the term.

16

Family night

N05 • T01 • C05

Hosting, the stop-motion films screened, and something cooked for someone who will be glad of it.

Cards not placed in either sequence are the bank, and they are not lesser — several are among the strongest in the deck. Swap them in when the kitchen is unavailable (M01, M02, M03, T02, T04, A10, E01 all run in an ordinary classroom), when the weather or the term throws a session, or when a card connects to something the children are already studying.

Equipment, materials and what to beg for

WORTH BUYING FIRST

- Two or three digital scales reading grams — the highest-value purchase in the whole course.
- Four instant-read thermometers, so that children can discover they disagree.
- Hand lenses, one per pair, and a clip-on macro lens for a tablet if the budget stretches.
- Nylon or serrated children's knives for grades 2–3, and two small chef's knives for grades 4–5.
- Clear glass jars and cups in quantity — almost every observation improves when you can see the side.
- Two identical sheet pans, so paired batches bake under identical conditions.

ART MATERIALS THE DECK ACTUALLY NEEDS

- Alum (spice aisle), white cotton squares cut from old sheets, string and rubber bands.
- Block printing ink or thick tempera, a brayer, a flat tray, newsprint and good paper.
- Graphite pencils in two grades, kneaded erasers, a small watercolor set, heavy paper.
- Tweezers and a small offset spatula for plating, which cost little and change the work.

FREE, AND WORTH ARRANGING

- A wall for the word bank and a second for the food map. Both fill over the term.
- A window with daylight for the photography card, and a dark corner for the sound card.
- Old sheets, jars, cardboard, boxes. Ask staff, the school office, and put a donation bin by the door — never a note home naming what a family should send. A request that arrives in a child's bag is a request a family can be seen to fail.
- The school's own kitchen staff, as visiting experts. Ask early, give notice, let them choose the topic.

The five-minute check, after every session

Ten questions. Answer them honestly and alone. The first one is the one that matters most.

1. Did I comment on what any child ate today? Including approvingly?
2. Did every child hold a named job, and did the sensory reporter role rotate?
3. Did the youngest children do something that mattered, or did they watch?
4. Was every ingredient named out loud before it went in?
5. Did anyone — me, an aide, a volunteer, a visiting parent — say “just try it”?
6. Did I redirect a “yuck” to a description, calmly, the first time it happened?
7. Did anything about bodies, weight, healthy or junk enter the room? From me?
8. Were components served separately, and was there a no-thank-you plate at every setting?
9. Has anything happened over recent weeks that the coordinator should know — distress escalating rather than settling, or a child starting to avoid the class? Not: did anyone eat less than last week.
10. Did the room feel like somewhere a worried eater would come back to next week?

A note on the evidence. The guidance on pressure, praise and food-neutral teaching is strong and consistent. The named feeding-therapy hierarchies are widely used in practice but less well evidenced than their ubiquity suggests. The reward question is genuinely contested and this guide takes the cautious side of it for group settings. And there is very little research on mixed-age primary cooking classes specifically — that section is assembled from adjacent evidence and established program models, and should be read as informed practice rather than as established fact.

The forty-six cards

Seven strands. The companion deck holds one card per page in this order.

Senses & Self

N01 The Taster's Alphabet

Building a sensory vocabulary the whole class shares

N02 Eight Ways to Meet a Food

The class agreement, taught to everybody on day one

N03 Listen to It First

Sound as a food property, and the easiest door in

N04 Three Cups, One Difference

The triangle test — how professionals ask whether a difference is real

N05 A Table Where Everyone Fits

Hosting, asking for what you need, and minding your own plate

Science

S01 The Cucumber Weeps

Osmosis — water moving where you did not put it

S02 Pulling Colors Apart

Capillary action and paper chromatography

S03 Three Ways to Set a Liquid

Gelatin, agar and pectin — the gels do not behave alike

S04 The Apple Goes Brown

Enzymatic browning, and four ways to stop it

S05 Your Tongue Is Not My Tongue

How plants defend themselves, and why people taste them differently

S06 Four Families of Color

What heat and acid do to the pigments in plants

S07 Closer Than You Have Ever Looked

Magnification, crystals and the structure of ordinary things

S08 The Cracker Turns Sweet

Digestion as a system — mechanical and chemical

Art

A01 The Color in the Peel

Natural dyes, mordants and colorfastness

A02 A Color Wheel You Can Eat

Primary, secondary and complementary color, in produce

A03 The Okra Is a Stamp

Relief printmaking, repeat pattern and the edition

A04 Draw It Before You Know It

Observational and botanical illustration

A05 The Plate Is a Frame

Composition — thirds, odd numbers, negative space, line

A06 Where Is the Light Coming From?

Photographing food — direction, diffusion, angle

A07 Dough Is a Material

Sculpture — form, volume and the thing that holds its shape

A08 Round and Round and Round

Radial symmetry, repeat units and pattern

A09 The Kitchen Orchestra

Sound, rhythm and a score anyone can read

A10 What Shall We Call It?

Naming, typography and the identity of a dish

Engineering

E01 A Tool for This Hand

Designing for a real person, not an average one

E02 Which Shape Holds the Sauce?

Pasta geometry, surface texture and a measurable test

E03 The Lunchbox Problem

Structure, moisture and a sandwich that survives until noon

E04 What Kneading Actually Does

Dough as a material, and a test you can hold up to the light

E05 Making Water Clear Is Not Making It Safe

Filtration, and the difference between clean and clear

E06 Design This Kitchen

Workflow, stations and how far a person has to walk

Technology

T01 Twelve Pictures a Second

Stop-motion animation of a recipe

T02 Ask, Then Stop Talking

Recording an interview about a food that matters to someone

T03 The Cold Chain

Thermometers, data logging, and the gap food must not sit in

T04 Make the Book

Layout, editing and publishing the class cookbook

T05 Before Anyone Owned a Thermometer

How people cooked precisely without precise instruments

Math

M01 Count the Spirals

Fibonacci numbers, growing things, and a pattern nobody planned

M02 Cooking for Two Hundred

Scaling up, division with remainders, and what a tray holds

M03 The Potato in the Jug

Volume of a lumpy object, by displacement

M04 Watching Something Cool Down

Line graphs, rate of change, and a curve that flattens

M05 The Class Vote

Anonymous data, fair comparison, and what a number can honestly claim

M06 A Pinch, a Handful, a Gram

Non-standard and standard units, and why anyone bothered

Table & Community

C01 The Dish That Traveled

A recipe with a person attached to it

C02 One Grain, Many Kitchens

The same staple, cooked completely differently

C03 The Part You Were Going to Throw Away

Whole-ingredient cooking, stock, and what waste actually is

C04 Everyone Who Touched It

The people between a field and a plate

C05 Cook It for Someone Else

A real recipient, a real delivery

C06 How the World Eats

Utensils, table shapes and the rules nobody wrote down

Where this comes from

The design decisions in this guide are not inventions. These are the sources behind them.

FEEDING AND EXPOSURE

- SOS Approach to Feeding — the “Steps to Eating” hierarchy (Toomey). The 32-step, six-band ladder this course’s wheel is adapted from.
- Fishbein et al., “Food Chaining: a systematic approach for the treatment of children with feeding aversion”, *Nutrition in Clinical Practice*, 2006 — sequencing by color, texture, temperature and shape.
- NHS guidance on sensory approaches to eating (Humber; Derbyshire Local Offer) — the child-facing seven-step version, and the instruction never to force.

PRESSURE, PRAISE AND REWARDS

- Galloway, Fiorito, Francis & Birch, “‘Finish your soup’: counterproductive effects of pressuring children to eat on intake and affect”, *Appetite*, 2006.
- Ellyn Satter Institute — Division of Responsibility, and the child-care handout: children eat less well, not better, when pressured; “no thank you bites” slow learning.
- CNTW NHS — “praise can be interpreted as pressure and can be counterproductive.”
- Contested, and worth knowing: an RCT in Public Health Nutrition found sticker rewards plus repeated exposure increased vegetable liking in 4–6-year-olds without undermining intrinsic motivation. That evidence comes from typically-developing children in a one-to-one home protocol, which is why this course still says no rewards in a group.

ARFID AND SCHOOLS

- CNTW NHS — “How can schools support a child with ARFID?” and “Helping your child add new foods”.

- Grender, “Understanding and Preventing Maltreatment with ARFID: a guide for teachers and families”, Council for Exceptional Children — names coercive feeding, food-linked punishment and public commentary on eating as maltreatment.
- It’s Not Picky Eating — PFD & ARFID school toolkit; Feeding Matters — school-based accommodations.
- NEDA — Understanding ARFID in children and teens.

FOOD-NEUTRAL TEACHING

- Teach Food First (BC) — Guiding Principles for Educators: avoid sorting foods into healthy and unhealthy; children are less likely to try foods labeled “healthy”.
- Equip — Guide for Educators; National Alliance for Eating Disorders — Tips for Removing Morality from Food.

SENSORY FOOD EDUCATION

- Sapere Association — the sensory education method.
- Mustonen & Tuorila, Food Quality and Preference — sensory education decreased food neophobia in 8–12-year-olds.
- Public Health Nutrition — sensory-based food education in early childhood settings; cross-sectional, and it found sensory groups did not show the produce-avoidance drift seen in reference groups.
- British Nutrition Foundation — sensory vocabulary poster (trim the disgust words before use).

RUNNING THE ROOM

- FoodCorps — Taste Test Guidebook: anonymous bean-drop and dot voting, class-level results only.
- NSW Health — Kids in the Kitchen manual: stations of five to six, different recipes per station, small taster portions across stations.
- CDC and FARE — food allergy management in schools; design the menu around the roster.